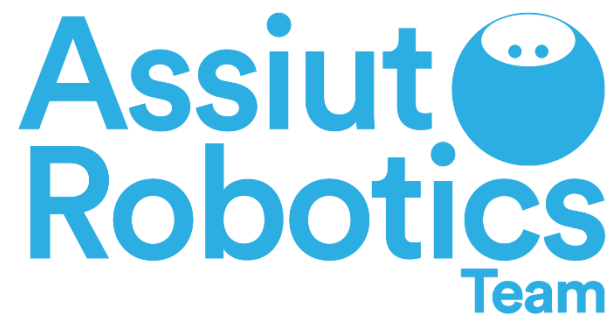


Arduino basics



Weak #03

Main objectives:

- Introduction to DC motors & motor drivers
- Be familiar with servo motor
- Interface with keypad

Study material:

- [Arduino course](#) -> 15, 28, 14, 27
(please follow the sequence on watching the video)
- [Arduino course](#) -> 29 & 30

1st_Task:

Control two motors with motor driver (L293D recommended), Create 4 functions (forward, backward, turn_right, turn_left), and attach 4 pushbuttons for each movement to control the movement direction,

BONUS: Create difference code to control the motor speed using ultrasonic reading,

The main goal is to make the motor speed related to the distance, the higher the distance the faster the motor rotates

Hint: search for P controller or PID controller

2nd_Task:

Create a digital lock using a servo motor, keypad, and LCD.

The target of this application is to develop a secure system with a password, we can insert the password using the keypad, and if the inserted password is correct the servo motor will rotate 90 degrees, and if it is wrong the servo motor will reset (rotate to 0 degree) and buzzer alarm will be triggering for 3 sec

Hint: password must appear on the LCD as stars (*)